Statistical Inference

하지운

Sungkyunkwan University

통계학과

Contents

1. Motivation
2. Statistical Inference
   - Convergence in Probability
   - Convergence in Distribution
   - Maximum Likelihood Estimator
   - Rao–Cramér Lower Bound and Efficiency
   - Sufficient Statistics
   - Completeness and Uniqueness
3. Appendix: Proofs
4. References

Motivation

Motivation

1. How can we learn about an unknown population from a limited amount of data?
2. How can we estimate unknown parameters using observed samples?
3. How can we determine whether an estimator is reliable?

An unknown probability

A delivery service makes many deliveries, but its true probability p of arriving on time is unknown.

Suppose we observe **10 deliveries**: **8 on time** and **2 late**.

We might estimate the on-time probability as 0.8.
However, several questions remain:

- Is 0.8 a reasonable estimate of p?
- Could the true probability be different?
- How reliable is our estimate?

These are fundamental questions in **Statistical Inference**.

Example

A delivery service makes 10 independent deliveries, of which 8 arrive on time. Compare $p = 0.5$ and $p = 0.8$ under the binomial model.

Solution. The probability of observing exactly eight on-time deliveries is

$$P_p(X = 8) = \binom{10}{8} p^8 (1-p)^2.$$

Assumed on-time rate	$p = 0.5$	$p = 0.8$
Probability of 8 on time	0.044	0.302

The observation is approximately **6.9 times as likely** under $p = 0.8$.

This suggests that $p = 0.8$ explains the observed data better, but it does not establish the true value of p.

From Data to Estimation

Statistical inference begins with observed data and attempts to draw conclusions about unknown population parameters.

For the delivery example, one possible approach is to maximize the likelihood function:

$$L(p) = \binom{10}{8} p^8 (1-p)^2.$$

The maximizing value is

$$\widehat{p} = \arg\max_{0 \le p \le 1} L(p) = 0.8.$$

This is an example of **Maximum Likelihood Estimation (MLE)**.

However, obtaining an estimate is not enough. We must also investigate its statistical properties.

Example

A cafe records 2, 5, 3, and 4 orders in four equal time periods. Under a Poisson model, estimate the average number of orders per period.

Solution. The sample mean gives an estimate of the unknown parameter λ:

$$\widehat{\lambda} = \frac{2+5+3+4}{4} = 3.5.$$

The delivery example concerns an unknown **probability**, whereas the cafe example concerns an unknown **average count**.

In both cases, we use a sample to estimate a population parameter. The next question is whether these estimators have desirable statistical properties.

Why Study Statistical Inference?

Statistical inference provides methods for estimating unknown parameters and evaluating the quality of estimators.

- **Consistency:** Does the estimator approach the true parameter as the sample size increases?
- **Limiting Distribution:** How does the estimator behave for large samples?
- **Maximum Likelihood Estimation:** How can we construct estimators using likelihood?
- **Fisher Information and CRLB:** How precisely can a parameter be estimated under suitable conditions?
- **Sufficiency and Completeness:** How can we summarize the information contained in a sample?
- **MVUE:** How can we identify an unbiased estimator with minimum variance?

Convergence in Probability

Definition (Convergence in probability)

A sequence X_n converges in probability to X if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} P(|X_n - X| \geq \varepsilon) = 0.$$

We write $X_n \xrightarrow{P} X$.

Theorem (Markov's inequality)

For a nonnegative random variable Y with finite expectation and $c > 0$,

$$P(Y \geq c) \leq \frac{\mathrm{E}[Y]}{c}.$$

Theorem (Chebyshev's inequality)

Let X have finite mean $\mu = \mathrm{E}[X]$ and finite variance $\sigma^2 = \mathrm{Var}(X)$. For every $\varepsilon > 0$,

$$P(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Apply Markov's inequality to the nonnegative random variable $Y = (X - \mu)^2$ and use $\mathrm{E}[Y] = \sigma^2$:

$$P(|X - \mu| \geq \varepsilon) = P(Y \geq \varepsilon^2) \leq \frac{\mathrm{E}[Y]}{\varepsilon^2} = \frac{\sigma^2}{\varepsilon^2}. \quad \square$$

Theorem (Weak Law of Large Numbers)

Let $X_1, \ldots, X_n$ be i.i.d. with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Then the sample mean $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$ satisfies $\bar{X}_n \xrightarrow{P} \mu$.

Proof. Since the observations are independent,

$$\mathrm{E}[\bar{X}_n] = \mu, \qquad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

For every $\varepsilon > 0$, Chebyshev's inequality gives

$$P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

This is convergence in probability. □

Example

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$ and $\widehat{p} = \bar{X}_n$. Show $\widehat{p} \xrightarrow{P} p$.

Solution. Here $\text{E}[\widehat{p}] = p$ and $\text{Var}(\widehat{p}) = p(1-p)/n$. Thus,

$$P(|\widehat{p} - p| \geq \varepsilon) \leq \frac{p(1-p)}{n\varepsilon^2} \to 0.$$

Alternatively, $\widehat{p} \xrightarrow{P} p$ follows directly from the WLLN.

Theorem (Linearity of convergence in probability)

Suppose $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y$. Then $X_n + Y_n \xrightarrow{P} X + Y$. Also, for every constant $a \in \mathbb{R}$, $aX_n \xrightarrow{P} aX$.

Proof. Put $A_n = |X_n - X|$ and $B_n = |Y_n - Y|$. By the triangle inequality and union bound,

$$P(|(X_n + Y_n) - (X + Y)| \geq \varepsilon) \leq P(A_n \geq \varepsilon/2) + P(B_n \geq \varepsilon/2).$$

Both probabilities tend to zero. If $a \neq 0$, then

$$P(|aX_n - aX| \geq \varepsilon) = P(|X_n - X| \geq \varepsilon/|a|) \longrightarrow 0.$$

For $a = 0$, the assertion is immediate. □

Theorem (Continuous mapping for convergence in probability)

If $X_n \xrightarrow{P} a$ for a constant a, and g is continuous at a, then $g(X_n) \xrightarrow{P} g(a)$.

Proof. Given $\varepsilon > 0$, continuity at a provides $\delta > 0$ such that $|x - a| < \delta$ implies $|g(x) - g(a)| < \varepsilon$. Hence

$$\{|g(X_n) - g(a)| \geq \varepsilon\} \subseteq \{|X_n - a| \geq \delta\}.$$

Consequently,

$$P(|g(X_n) - g(a)| \geq \varepsilon) \leq P(|X_n - a| \geq \delta) \longrightarrow 0.$$

Thus $g(X_n) \xrightarrow{P} g(a)$. □

Theorem (Product of convergent random variables)

If $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y$, then $X_n Y_n \xrightarrow{P} XY$.

Proof. By linearity, $X_n + Y_n \xrightarrow{P} X + Y$ and $X_n - Y_n \xrightarrow{P} X - Y$. The continuous mapping theorem for $g(x) = x^2$ gives convergence of their squares. Now use

$$X_n Y_n = \frac{1}{2}\left\{(X_n + Y_n)^2 - X_n^2 - Y_n^2\right\}.$$

Applying linearity once more yields

$$X_n Y_n \xrightarrow{P} \frac{1}{2}\left\{(X + Y)^2 - X^2 - Y^2\right\} = XY.$$

□

Definition (Consistent estimator)

For a parameter $\theta \in \Theta$, a sequence of estimators T_n is consistent if

$$T_n \xrightarrow{P} \theta \qquad \text{for every } \theta \in \Theta.$$

Consistency concerns what happens as the sample size increases. It does not assert that T_n is unbiased for any fixed n.

Example

Let X_i be i.i.d. with finite mean μ and variance σ^2. Determine whether $\bar{X}_n$ is consistent for μ.

Solution. By the WLLN, $\bar{X}_n \xrightarrow{P} \mu$; therefore $\bar{X}_n$ is consistent.

Unbiasedness Versus Consistency

- **Unbiasedness** means $\mathrm{E}_\theta[T_n] = \theta$ for the specified sample size.
- **Consistency** means $P_\theta(|T_n - \theta| \geq \varepsilon) \to 0$ as $n \to \infty$.

$$\mathrm{MSE}_\theta(T_n) = \mathrm{Var}_\theta(T_n) + \{\mathrm{E}_\theta[T_n] - \theta\}^2.$$

An unbiased estimator may have large variance, and a biased estimator may nonetheless be consistent. If MSE tends to zero, consistency follows by Markov's inequality.

How to Check Consistency

There are three common ways to establish consistency:

1. Express the estimator through sample averages and use the WLLN.
2. Show that $\mathrm{MSE}(T_n) \to 0$.
3. If neither method is convenient, evaluate $P_\theta(|T_n - \theta| \geq \varepsilon)$ directly.

Example

Let X_i be i.i.d. with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Show that $S_n^2 = (n-1)^{-1} \sum_i (X_i - \bar{X}_n)^2$ is consistent for σ^2.

Solution. By the WLLN, $n^{-1} \sum_i X_i^2 \xrightarrow{P} \mathrm{E}[X_1^2]$ and $\bar{X}_n \xrightarrow{P} \mu$. Hence

$$S_n^2 = \frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^{n} X_i^2 - \bar{X}_n^2 \right) \xrightarrow{P} \mathrm{E}[X_1^2] - \mu^2 = \sigma^2.$$

Example

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$ and $\widehat{\theta} = \bar{X}_n$. Show that $\widehat{\theta} \xrightarrow{P} \theta$.

Solution. Since $\text{E}_\theta[\bar{X}_n] = \theta$ and $\text{Var}_\theta(\bar{X}_n) = \theta/n$,

$$\text{MSE}_\theta(\widehat{\theta}) = \frac{\theta}{n} \longrightarrow 0.$$

Therefore $\widehat{\theta} \xrightarrow{P} \theta$.

Example

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Unif}(0, \theta)$ and $Y_n = X_{(n)} = \max_i X_i$. Show that $Y_n \xrightarrow{P} \theta$.

Solution. For $0 < \varepsilon < \theta$, since $Y_n \leq \theta$,

$$P_\theta(|Y_n - \theta| \geq \varepsilon) = P_\theta(Y_n \leq \theta - \varepsilon) = \left(1 - \frac{\varepsilon}{\theta}\right)^n \to 0.$$

Thus $Y_n \xrightarrow{P} \theta$.

$\text{E}_\theta[Y_n] = n\theta/(n+1) \neq \theta$. The estimator is biased at every finite sample size but consistent.

Convergence in Distribution

Motivation: Convergence in Distribution

Suppose $U_n \sim N(0, 1 + 1/n)$ and $U \sim N(0, 1)$ are independent.

- Their distributions approach each other as n increases.
- But $U_n - U \sim N(0, 2 + 1/n)$, so U_n does not converge in probability to this independent U.

Definition (Convergence in distribution)

Let F_n be the CDF of X_n and F that of X. We write $X_n \xrightarrow{d} X$ if

$$F_n(x) \longrightarrow F(x) \qquad \text{at every continuity point of } F.$$

The condition is imposed only at continuity points of the limiting CDF; at jumps, pointwise CDF convergence is not required.

Example

Let $X_n \equiv 1/n$ and $X \equiv 0$ be degenerate random variables. Check whether $X_n \xrightarrow{d} X$.

Solution.

$$F_n(x) = \mathbf{1}_{\{x \geq 1/n\}}, \qquad F(x) = \mathbf{1}_{\{x \geq 0\}}.$$

For every $x \neq 0$, $F_n(x) \to F(x)$; nevertheless $F_n(0) = 0 \neq F(0)$.

Example

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Unif}(0, \theta)$, $Y_n = X_{(n)}$, and $Z_n = n(\theta - Y_n)$. Find the limiting distribution of Z_n.

Solution. For $t \geq 0$ and sufficiently large n,

$$P(Z_n \leq t) = 1 - P(Y_n < \theta - t/n) = 1 - \left(1 - \frac{t}{n\theta}\right)^n.$$

Therefore

$$Z_n \xrightarrow{d} \text{Exp}(1/\theta).$$

Although $Y_n \xrightarrow{P} \theta$, rescaling the estimation error by n produces a nondegenerate exponential limiting distribution.

Theorem (Basic relationship)

If $X_n \xrightarrow{P} X$, then $X_n \xrightarrow{d} X$.

Theorem (Constant limit)

If $X_n \xrightarrow{d} c$ for a constant c, then $X_n \xrightarrow{P} c$.

In general, convergence in distribution is weaker than convergence in probability. When the limiting random variable is constant, the two notions are equivalent.

Theorem (Continuous mapping)

If $X_n \xrightarrow{d} X$ and g is continuous, then $g(X_n) \xrightarrow{d} g(X)$.

Theorem (Slutsky)

If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{P} a$, then

$$X_n + Y_n \xrightarrow{d} X + a, \quad X_n Y_n \xrightarrow{d} aX, \quad X_n / Y_n \xrightarrow{d} X/a \ (a \neq 0).$$

Slutsky's theorem allows unknown variance parameters to be replaced by consistent estimators when deriving limiting distributions.

Moment Generating Functions

Under conditions ensuring the uniqueness and convergence of moment generating functions, one can establish convergence in distribution by showing

$$M_{X_n}(t) = \mathrm{E}[e^{tX_n}] \longrightarrow M_X(t)$$

for t in an open interval containing 0.

Theorem (Central Limit Theorem)

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathrm{E}[X_i] = \mu$ and $0 < \mathrm{Var}(X_i) = \sigma^2 < \infty$. Then

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1).$$

Proof. Let $Z_i = (X_i - \mu)/\sigma$. Since $\mathrm{E}[Z_i] = 0$ and $\mathrm{E}[Z_i^2] = 1$, the characteristic function satisfies

$$\varphi_Z(t) = 1 - \frac{t^2}{2} + o(t^2) \qquad (t \to 0).$$

By independence, the characteristic function of $n^{-1/2} \sum_i Z_i$ is

$$\left\{\varphi_Z(t/\sqrt{n})\right\}^n = \left\{1 - \frac{t^2}{2n} + o(n^{-1})\right\}^n \longrightarrow e^{-t^2/2}.$$

Lévy's continuity theorem gives the asserted weak convergence. □

Example

Let X_i be i.i.d. with mean μ and finite variance $\sigma^2 > 0$. Let $S_n^2 = (n-1)^{-1}\sum_i (X_i - \bar{X}_n)^2$. Find the limiting distribution of the studentized sample mean.

Solution. By the CLT, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0,1)$; by consistency of sample variance, $S_n/\sigma \xrightarrow{P} 1$.

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \xrightarrow{d} N(0,1).$$

Slutsky's theorem justifies replacing σ by S_n. This is the basis for a large-sample confidence interval for μ.

Theorem (Delta method)

Suppose $\sqrt{n}(X_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ and g is differentiable at θ, with $g'(\theta) \neq 0$. Then

$$\sqrt{n}\{g(X_n) - g(\theta)\} \xrightarrow{d} N\Big(0, \sigma^2[g'(\theta)]^2\Big).$$

Proof. The assumed convergence implies $X_n \xrightarrow{P} \theta$. Differentiability gives

$$g(X_n) - g(\theta) = (X_n - \theta)\{g'(\theta) + r(X_n)\}, \qquad r(X_n) \xrightarrow{P} 0.$$

Multiply by $\sqrt{n}$; Slutsky's theorem implies

$$\sqrt{n}\{g(X_n) - g(\theta)\} \xrightarrow{d} g'(\theta)Z, \qquad Z \sim N(0, \sigma^2).$$

The limit has the stated variance. □

Example

Let $Y \sim \text{Binomial}(n, p)$ and $\widehat{p} = Y/n$ with $0 < p < 1$. Find the limiting distribution of the logit estimator $g(\widehat{p})$, where $g(p) = \log\{p/(1-p)\}$.

Solution. The CLT gives $\sqrt{n}(\widehat{p} - p) \xrightarrow{d} N(0, p(1-p))$. For the logit $g(p) = \log\{p/(1-p)\}$, we have $g'(p) = 1/[p(1-p)]$. Thus

$$\sqrt{n}\{g(\widehat{p}) - g(p)\} \xrightarrow{d} N\left(0, \frac{1}{p(1-p)}\right).$$

The derivative amplifies uncertainty when p is near 0 or 1. The logit is defined whenever $0 < \widehat{p} < 1$.

Example

Let X_i be i.i.d. with $\mathrm{E}[X_i] = \mu \neq 0$ and $\mathrm{Var}(X_i) = \sigma^2 < \infty$. Find the limiting distribution of $1/\bar{X}_n$.

Solution.

$$\sqrt{n}\left(\frac{1}{\bar{X}_n} - \frac{1}{\mu}\right) \xrightarrow{d} N\left(0, \frac{\sigma^2}{\mu^4}\right).$$

By Slutsky's theorem, replacing μ, σ with $\bar{X}_n, S_n$ leads to

$$\frac{\sqrt{n}(1/\bar{X}_n - 1/\mu)}{S_n/\bar{X}_n^2} \xrightarrow{d} N(0,1).$$

Maximum Likelihood Estimator

Definition (Statistic)

A statistic $T = T(X_1, \ldots, X_n)$ is a function of a random sample. Its observed value is $t = T(x_1, \ldots, x_n)$.

Definition (Estimator and estimate)

When T is used to estimate an unknown model parameter θ, it is an estimator; its realized value t is an estimate.

Definition (Likelihood)

For an observed i.i.d. sample $x = (x_1, \ldots, x_n)$ from $f(x \mid \theta)$, define

$$L(\theta \mid x) = \prod_{i=1}^{n} f(x_i \mid \theta), \qquad \ell(\theta) = \log L(\theta \mid x).$$

The same density or mass function is now viewed as a function of θ with the data fixed. Likelihood is not generally a probability density over the parameter.

Interpretation of Likelihood Function

Let $X_1, \ldots, X_{100} \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$, and suppose 93 successes are observed. Compare $\theta = 0.1$ with $\theta = 0.9$.

$$L(\theta \mid x) \propto \theta^{93}(1-\theta)^7.$$

The likelihood ratio satisfies

$$\frac{L(0.9 \mid x)}{L(0.1 \mid x)} = 9^{86} > 1.$$

For this observed sample, $\theta = 0.9$ gives a much larger likelihood than $\theta = 0.1$.

Definition (MLE)

The maximum likelihood estimator is any parameter value maximizing the likelihood over the parameter space Θ:

$$\widehat{\theta} \in \arg\max_{\theta \in \Theta} L(\theta \mid x) = \arg\max_{\theta \in \Theta} \ell(\theta).$$

The logarithm is strictly increasing, so maximizing the log-likelihood is equivalent and usually easier. Existence and uniqueness of the maximum still need to be checked.

Theorem (Likelihood comparison)

Let $X_i \overset{\text{iid}}{\sim} f(x \mid \theta_0)$. Assume identifiability, common support, and integrable log-likelihood ratios. For every fixed $\theta \neq \theta_0$,

$$P_{\theta_0}\{L_n(\theta_0) > L_n(\theta)\} \longrightarrow 1 \qquad (\theta \neq \theta_0).$$

Proof. By the WLLN,

$$\frac{1}{n} \log \frac{L_n(\theta)}{L_n(\theta_0)} \xrightarrow{P} \mathrm{E}_{\theta_0}\left[\log \frac{f(X \mid \theta)}{f(X \mid \theta_0)}\right].$$

By strict Jensen's inequality and identifiability,

$$\mathrm{E}_{\theta_0}\left[\log \frac{f(X \mid \theta)}{f(X \mid \theta_0)}\right] < \log \mathrm{E}_{\theta_0}\left[\frac{f(X \mid \theta)}{f(X \mid \theta_0)}\right] = 0.$$

Therefore $P_{\theta_0}[L_n(\theta_0) > L_n(\theta)] \to 1$. □

Properties of the MLE

- An MLE must belong to the specified parameter space Θ.
- A maximizer can be on the boundary, even if the score does not vanish there.
- An MLE may be nonunique or may fail to exist.

Example

Suppose $X_i \sim N(\theta, 1)$ and $\bar{x} = -0.1$. Compare the MLE under $\theta \geq 0$ versus $\theta > 0$.

Solution. For $\theta \geq 0$, $\widehat{\theta} = 0$. For $\theta > 0$, the likelihood supremum is approached as $\theta \downarrow 0$ but never attained.

How to Find an MLE

1. Write down $L(\theta \mid x)$ and the parameter space.
2. Compute the log-likelihood $\ell(\theta)$.
3. Solve $\ell'(\theta) = 0$ at differentiable interior candidates.
4. Check that each candidate is a *global* maximizer, including parameter boundaries.

Theorem (Strictly concave log-likelihood)

Let Θ be an interval and suppose ℓ is differentiable and strictly concave on Θ. If an interior point $\widehat{\theta} \in \Theta$ satisfies $\ell'(\widehat{\theta}) = 0$, then $\widehat{\theta}$ is the unique global maximizer of ℓ on Θ.

Proof. The tangent line to a differentiable strictly concave function strictly exceeds its value at any other point. Therefore, for $\theta \neq \widehat{\theta}$,

$$\ell(\theta) < \ell(\widehat{\theta}) + \ell'(\widehat{\theta})(\theta - \widehat{\theta}) = \ell(\widehat{\theta}).$$

Hence the stationary point is the unique MLE. □

Strict concavity alone does not guarantee that a stationary point exists; boundaries still matter.

Theorem (An interior maximum exists)

Let $\Theta = (a, b)$ be an open interval (possibly unbounded). Assume ℓ is twice differentiable, $\ell''(\theta) < 0$ on Θ, and

$$\lim_{\theta \downarrow a} \ell(\theta) = \lim_{\theta \uparrow b} \ell(\theta) = -\infty.$$

Then $\ell'(\widehat{\theta}) = 0$ has exactly one solution in Θ, which is the unique MLE.

Proof. The boundary limits force a maximum to occur away from both ends of Θ. At an interior maximum, the derivative is zero. Since $\ell'' < 0$, the derivative is strictly decreasing; it can vanish at most once. Thus the maximizer exists and is unique. □

Example

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ with $0 \le \theta \le 1$. Find the MLE of θ. Put $S = \sum_i X_i$.

Solution.

$$\ell(\theta) = S \log \theta + (n - S) \log(1 - \theta).$$

For $0 < S < n$,

$$\ell'(\theta) = \frac{S}{\theta} - \frac{n - S}{1 - \theta} = 0 \quad \Longrightarrow \quad \widehat{\theta} = S/n = \bar{X}_n.$$

Since $\ell''(\theta) = -S/\theta^2 - (n - S)/(1 - \theta)^2 < 0$, this is the unique maximum. The all-zero and all-one samples attain their maxima at 0 and 1.

Example

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Gamma}(1, \theta)$ with *scale* parameter $\theta > 0$. Find the MLE of θ.

Solution.

$$\ell(\theta) = -n \log \theta - \frac{\sum_i X_i}{\theta}.$$

The derivative is

$$\ell'(\theta) = \frac{n(\bar{X}_n - \theta)}{\theta^2}.$$

The derivative changes from positive to negative at $\theta = \bar{X}_n$, so $\widehat{\theta} = \bar{X}_n$.

Example

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Unif}(0, \theta)$, $\theta > 0$. Find the MLE of θ.

Solution.

$$L(\theta \mid x) = \theta^{-n} \mathbf{1}_{\{\theta \geq x_{(n)}\}}.$$

When $\theta < x_{(n)}$ the likelihood is zero; on $\theta \geq x_{(n)}$, it decreases with θ.

$$\widehat{\theta} = X_{(n)}.$$

Here the support depends on θ. Differentiating without the support indicator would miss the maximizer.

Joint Maximum Likelihood Estimation

For a vector parameter $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_k)$, the MLE maximizes the *joint* likelihood, not one parameter's likelihood in isolation:

$$\widehat{\boldsymbol{\theta}} \in \arg\max_{\boldsymbol{\theta} \in \Theta} \ell(\boldsymbol{\theta}).$$

At an interior differentiable maximum, all partial score equations must hold:

$$\frac{\partial \ell}{\partial \theta_j}(\widehat{\boldsymbol{\theta}}) = 0, \qquad j = 1, \ldots, k.$$

Solving these equations gives candidates, not an automatic maximum.

The next normal-model example first optimizes μ for fixed σ^2 and then optimizes the remaining variance parameter.

Example

Let $X_1, \ldots, X_n$ be i.i.d. Laplace(μ, b) with $f(x \mid \mu, b) = (2b)^{-1} e^{-|x-\mu|/b}$, $\mu \in \mathbb{R}$, $b > 0$. Assume the observations are not all equal. Find both MLEs.

Solution. The joint log-likelihood is

$$\ell(\mu, b) = -n \log(2b) - \frac{1}{b} \sum_{i=1}^{n} |x_i - \mu|.$$

For fixed b, it is maximized by any sample median $\widehat{\mu}$. For fixed μ, differentiate with respect to b to obtain

$$\frac{\partial \ell}{\partial b} = -\frac{n}{b} + \frac{\sum_i |x_i - \mu|}{b^2} = 0.$$

Hence $\widehat{b} = n^{-1} \sum_i |x_i - \widehat{\mu}|$. For even n, the optimizing location may be nonunique.

Example

Let $X_1, \ldots, X_n$ be i.i.d. Gamma(α, β), with shape $\alpha > 0$ and scale $\beta > 0$. Find the likelihood equations.

Solution. Up to an additive constant,

$$\ell(\alpha, \beta) = (\alpha - 1) \sum_i \log x_i - \frac{\sum_i x_i}{\beta} - n\alpha \log \beta - n \log \Gamma(\alpha).$$

Differentiating in the scale parameter gives

$$\frac{\partial \ell}{\partial \beta} = \frac{\sum_i x_i}{\beta^2} - \frac{n\alpha}{\beta} = 0 \quad \Longrightarrow \quad \widehat{\beta} = \frac{\bar{X}_n}{\widehat{\alpha}}.$$

The shape parameter must also satisfy its score equation.

Example — Gamma MLE (Continued)

Write $\psi(\alpha) = \Gamma'(\alpha)/\Gamma(\alpha)$ (the digamma function). The other score equation is

$$\frac{1}{n}\frac{\partial \ell}{\partial \alpha} = \overline{\log X} - \log \beta - \psi(\alpha) = 0.$$

Substitute $\widehat{\beta} = \bar{X}_n/\widehat{\alpha}$:

$$\log \widehat{\alpha} - \psi(\widehat{\alpha}) = \log \bar{X}_n - \overline{\log X}.$$

In general this equation is solved numerically; then $\widehat{\beta} = \bar{X}_n/\widehat{\alpha}$. Finding score roots still requires checking that a finite maximizer exists.

Example

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ with both parameters unknown. First find the MLE of μ for fixed σ^2.

Solution.

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_i (X_i - \mu)^2.$$

Using $\sum_i (X_i - \mu)^2 = \sum_i (X_i - \bar{X}_n)^2 + n(\bar{X}_n - \mu)^2$, the likelihood is maximized in μ at $\widehat{\mu} = \bar{X}_n$.

Example

For an i.i.d. normal sample with $\widehat{\mu} = \bar{X}_n$, find the MLE of $v = \sigma^2$.

Solution.

$$\frac{d}{dv}\ell(\bar{X}_n, v) = -\frac{n}{2v} + \frac{\sum_i (X_i - \bar{X}_n)^2}{2v^2}.$$

For a nonconstant sample the maximizing root is

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2.$$

The variance MLE uses divisor n, unlike the unbiased sample variance with divisor $n-1$. The MLE need not be unbiased.

Example

Let $X_1, \ldots, X_n$ be i.i.d. with logistic location density

$$f(x \mid \theta) = \frac{e^{-(x-\theta)}}{\{1 + e^{-(x-\theta)}\}^2}, \qquad \theta \in \mathbb{R}.$$

Show that the MLE exists and is unique.

Solution. The score and its derivative are

$$\ell'(\theta) = n - 2\sum_{i=1}^{n} \frac{e^{\theta - x_i}}{1 + e^{\theta - x_i}}, \quad \ell''(\theta) = -2\sum_{i=1}^{n} \frac{e^{\theta - x_i}}{(1 + e^{\theta - x_i})^2} < 0.$$

Moreover, $\ell'(-\infty) = n$ and $\ell'(+\infty) = -n$. Thus the score crosses zero exactly once; its unique root is the MLE.

Theorem (Invariance property)

Let $\widehat{\theta}$ maximize $L(\theta)$ on Θ, and let $\eta = g(\theta)$. Define the induced likelihood $L^*(\eta) = \sup_{\theta:g(\theta)=\eta} L(\theta)$. Then $\widehat{\eta} = g(\widehat{\theta})$ maximizes L^* on $g(\Theta)$.

Proof. Put $\widehat{\eta} = g(\widehat{\theta})$. Since $\widehat{\theta}$ is a global maximizer, for every $\eta \in g(\Theta)$,

$$L^*(\eta) = \sup_{\theta:g(\theta)=\eta} L(\theta) \leq L(\widehat{\theta}).$$

The fiber $\{\theta : g(\theta) = \widehat{\eta}\}$ contains $\widehat{\theta}$, so

$$L^*(\widehat{\eta}) = L(\widehat{\theta}) \geq L^*(\eta).$$

Therefore $g(\widehat{\theta})$ is an MLE of η. □

Example

For $X_i \overset{\text{iid}}{\sim} \text{Gamma}(1, \theta)$, the MLE is $\widehat{\theta} = \bar{X}_n$. The probability of observing a value at most 2 is

$$g(\theta) = P_\theta(X_1 \leq 2) = 1 - e^{-2/\theta}.$$

Find the MLE of $P_\theta(X_1 \leq 2)$.

Solution. By invariance,

$$\widehat{P_\theta(X_1 \leq 2)} = 1 - e^{-2/\bar{X}_n}.$$

Theorem (Population likelihood comparison)

Suppose the true distribution is $f(x \mid \theta_0)$, the model is identifiable, the support is common, and the expectations exist. Then

$$\mathrm{E}_{\theta_0}[\log f(X \mid \theta)] \leq \mathrm{E}_{\theta_0}[\log f(X \mid \theta_0)]$$

for every θ, with equality only if $\theta = \theta_0$.

Proof. Using $\log t \leq t - 1$ for $t > 0$ and common support,

$$\mathrm{E}_{\theta_0}\left[\log \frac{f(X \mid \theta)}{f(X \mid \theta_0)}\right] \leq \mathrm{E}_{\theta_0}\left[\frac{f(X \mid \theta)}{f(X \mid \theta_0)} - 1\right] = 0.$$

Equality forces equal distributions, so identifiability gives $\theta = \theta_0$. □

What the Likelihood Comparison Proves

For fixed θ, the law of large numbers suggests

$$\frac{1}{n}\ell_n(\theta) \xrightarrow{P} \mathrm{E}_{\theta_0}[\log f(X \mid \theta)].$$

The theorem shows that the true parameter uniquely maximizes the *expected* log-likelihood.

Important distinction

Pointwise convergence for each fixed parameter is not, by itself, a proof that the maximizer over an infinite parameter set converges to θ_0. Consistency also requires control of how the sample likelihood behaves near its maximum.

This explains why the regularity and uniqueness assumptions in the MLE consistency theorem matter.

Theorem (Consistent likelihood-equation root)

Assume the likelihood comparison result holds for every fixed alternative, θ_0 is interior to an interval Θ, and ℓ_n is continuous and differentiable near θ_0. Then a sequence of roots $\widehat{\theta}_n$ of $\ell_n'(\theta) = 0$ can be chosen so that $\widehat{\theta}_n \xrightarrow{P} \theta_0$.

Proof. Fix $\varepsilon > 0$ with $\theta_0 \pm \varepsilon \in \Theta$. The previous theorem implies, with probability tending to one,

$$\ell_n(\theta_0) > \ell_n(\theta_0 - \varepsilon), \qquad \ell_n(\theta_0) > \ell_n(\theta_0 + \varepsilon).$$

On this event, the maximum of ℓ_n on the closed interval is attained in its interior. Differentiability gives a root $\widehat{\theta}_n$ in that interval; hence $P(|\widehat{\theta}_n - \theta_0| < \varepsilon) \to 1$. □

Example

Show consistency of the MLE in the following models:
$X_i \overset{\text{iid}}{\sim} \text{Gamma}(1, \theta_0)$ and $X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta_0)$.

Solution.

- If $X_i \overset{\text{iid}}{\sim} \text{Gamma}(1, \theta_0)$, the MLE is $\widehat{\theta} = \bar{X}_n$, and $\bar{X}_n \xrightarrow{P} \theta_0$.
- If $X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta_0)$, $0 < \theta_0 < 1$, the MLE is $\widehat{\theta} = \bar{X}_n$, and $\bar{X}_n \xrightarrow{P} \theta_0$.

Both results follow from the WLLN because each MLE equals the corresponding sample mean.

Rao–Cramér Lower Bound and Efficiency

Regularity Conditions

The score identities, Fisher information and CRLB require the following regularity conditions:

(R0) Identifiability of the distribution family.

(R1) Support does not depend on θ.

(R2) θ_0 is an interior point of Θ.

(R3) Required first and second derivatives exist.

(R4) Differentiation under the integral or sum is justified.

Theorem (Score identities)

Under the regularity conditions permitting differentiation under the integral or sum,

$$\mathrm{E}_\theta[\frac{d}{d\theta}\log f(X \mid \theta)] = 0.$$

Moreover,

$$\mathrm{E}_\theta[(\frac{d}{d\theta}\log f(X \mid \theta))^2] = -\mathrm{E}_\theta[\frac{d^2}{d\theta^2}\log f(X \mid \theta)].$$

Definition (Fisher information)

Under regularity conditions, the information in one observation is

$$I_1(\theta) = \mathrm{E}_\theta \left[\left\{ \frac{d}{d\theta} \log f(X \mid \theta) \right\}^2 \right] = -\mathrm{E}_\theta [\frac{d^2}{d\theta^2} \log f(X \mid \theta)].$$

For an i.i.d. sample, $I_n(\theta) = nI_1(\theta)$.

Information is larger when the likelihood changes more sharply around the true parameter. Independence makes sample information additive.

Theorem (Rao–Cramér Lower Bound)

Let $X_1, \ldots, X_n$ be i.i.d. from a regular one-parameter family with $0 < I_1(\theta) < \infty$. If Y is a statistic with finite variance, $\mathrm{E}_\theta[Y] = k(\theta)$, and differentiation under the integral or sum is valid, then

$$\mathrm{Var}_\theta(Y) \geq \frac{\{k'(\theta)\}^2}{nI_1(\theta)}.$$

For an unbiased estimator $\widehat{\theta}$ of θ,

$$\mathrm{Var}_\theta(\widehat{\theta}) \geq \frac{1}{nI_1(\theta)}.$$

Since the bias is zero, its MSE equals its variance.

Example

Let $X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$, $0 < \theta < 1$. Find its Fisher information and check whether $\bar{X}_n$ attains the CRLB.

Solution. The one-observation log-likelihood has

$$-\ell_1''(\theta) = \frac{X}{\theta^2} + \frac{1-X}{(1-\theta)^2}, \qquad I_1(\theta) = \frac{1}{\theta(1-\theta)}.$$

For $\widehat{\theta} = \bar{X}_n$,

$$\text{Var}_\theta(\widehat{\theta}) = \frac{\theta(1-\theta)}{n} = \frac{1}{nI_1(\theta)}.$$

Definition (Efficient estimator)

An unbiased estimator is efficient if its variance attains the CRLB.

Definition (Efficiency of an unbiased estimator)

$$e(\widehat{\theta}) = \frac{1}{nI_1(\theta)\,\mathrm{Var}_\theta(\widehat{\theta})} \leq 1.$$

Efficiency equal to one means that the unbiased estimator attains the CRLB for the given sample size.

Example

Let $X_i \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$ and $\widehat{\theta} = \bar{X}_n$. Determine $I_1(\theta)$ and whether $\bar{X}_n$ attains the CRLB.

Solution.

$$\ell_1''(\theta) = -\frac{X}{\theta^2}, \qquad I_1(\theta) = \frac{1}{\theta}.$$

Moreover,

$$\text{Var}_\theta(\bar{X}_n) = \frac{\theta}{n} = \frac{1}{nI_1(\theta)}.$$

Example

Let $X_i \overset{\text{iid}}{\sim} \text{Beta}(\theta, 1)$, $\theta > 0$, $n > 2$. Find the MLE and the Fisher information.

Solution. From $f(x \mid \theta) = \theta x^{\theta-1}$ for $0 < x < 1$,

$$\ell_n'(\theta) = \frac{n}{\theta} + \sum_{i=1}^{n} \log X_i, \qquad \ell_n''(\theta) = -\frac{n}{\theta^2}.$$

Consequently,

$$\widehat{\theta} = -\frac{n}{\sum_{i=1}^{n} \log X_i}, \qquad I_n(\theta) = \frac{n}{\theta^2}.$$

Example

For $X_i \overset{\text{iid}}{\sim} \text{Beta}(\theta, 1)$, let $\widehat{\theta} = -n/\sum_i \log X_i$. Construct an unbiased estimator of θ and compare its variance with the CRLB.

Solution. Continuing the Beta model, gamma inverse moments give

$$\text{E}_\theta[\widehat{\theta}] = \frac{n\theta}{n-1}, \qquad \text{Var}_\theta\left(\frac{n-1}{n}\widehat{\theta}\right) = \frac{\theta^2}{n-2}.$$

Thus $(n-1)\widehat{\theta}/n$ is unbiased, but

$$\text{Var}_\theta\left(\frac{n-1}{n}\widehat{\theta}\right) = \frac{\theta^2}{n-2} > \frac{\theta^2}{n} = \frac{1}{I_n(\theta)}.$$

Theorem (Asymptotic normality)

Let $\widehat{\theta}_n$ be a consistent interior MLE in a regular one-parameter i.i.d. model. Assume conditions (R0)–(R5), an interior score equation, and $0 < I_1(\theta_0) < \infty$. Then

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \xrightarrow{d} N\left(0, \frac{1}{I_1(\theta_0)}\right).$$

Theorem (One-step estimator)

Assume a regular one-parameter i.i.d. model with valid score CLT, observed-information convergence and local Taylor expansion. Let an initial estimator $\widehat{\theta}_n^{(0)}$ satisfy

$$\sqrt{n}(\widehat{\theta}_n^{(0)} - \theta) = O_P(1).$$

Apply one Newton update to the sample log-likelihood:

$$\widehat{\theta}_n^{(1)} = \widehat{\theta}_n^{(0)} - \frac{\ell_n'(\widehat{\theta}_n^{(0)})}{\ell_n''(\widehat{\theta}_n^{(0)})}.$$

When the denominator is nonzero with probability tending to one,

$$\sqrt{n}(\widehat{\theta}_n^{(1)} - \theta) \xrightarrow{d} N\left(0, \frac{1}{I_1(\theta)}\right).$$

Example

Let $X_1, \ldots, X_n$ be i.i.d. from the standard-scale logistic location family

$$f(x \mid \theta) = \frac{e^{\theta - x}}{(1 + e^{\theta - x})^2}, \qquad \theta \in \mathbb{R}.$$

Starting at $\widehat{\theta}_n^{(0)} = \bar{X}_n$, obtain a one-step estimate.

Solution. Differentiation of the sample log-likelihood gives

$$\ell_n'(\theta) = n - 2\sum_{i=1}^{n} \frac{e^{\theta - X_i}}{1 + e^{\theta - X_i}}.$$

The second derivative is strictly negative:

$$\ell_n''(\theta) = -2\sum_{i=1}^{n} \frac{e^{\theta - X_i}}{(1 + e^{\theta - X_i})^2}.$$

Example

For the logistic location model, use the derivatives of the preceding slide and the initial estimator $\bar{X}_n$.

Solution. A single Newton update yields

$$\widehat{\theta}_n^{(1)} = \bar{X}_n - \frac{n - 2\sum_{i=1}^n \frac{e^{\bar{X}_n - X_i}}{1+e^{\bar{X}_n - X_i}}}{-2\sum_{i=1}^n \frac{e^{\bar{X}_n - X_i}}{(1+e^{\bar{X}_n - X_i})^2}}.$$

For this logistic location family, $\mathrm{E}_\theta[X] = \theta$, $\mathrm{Var}_\theta(X) = \pi^2/3$, and $I_1(\theta) = 1/3$. Thus $\bar{X}_n$ is a root-n consistent initial estimator, and

$$\sqrt{n}(\widehat{\theta}_n^{(1)} - \theta) \xrightarrow{d} N(0, 3).$$

Example

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Unif}(0, \theta)$, where $\theta > 0$. We found $\widehat{\theta} = X_{(n)}$. Determine the limit of $n(\theta - \widehat{\theta})$.

Solution. For $t \geq 0$ and sufficiently large n,

$$P_\theta\{n(\theta - X_{(n)}) > t\} = \left(1 - \frac{t}{n\theta}\right)^n \longrightarrow e^{-t/\theta}.$$

Hence

$$n(\theta - \widehat{\theta}) \xrightarrow{d} \text{Exp}(1/\theta).$$

The model's support depends on θ, so the regular $\sqrt{n}$-normal MLE theorem does not apply. This is the same extreme-value limit introduced earlier, now interpreted for the MLE.

Example

Find the limiting distribution of the MLE for i.i.d. Poisson(θ) and Beta$(\theta, 1)$ samples.

Solution. For $X_i \overset{\text{iid}}{\sim} \text{Poisson}(\theta)$, $\widehat{\theta} = \bar{X}_n$ and $I_1(\theta) = 1/\theta$. Therefore

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} N(0, \theta).$$

For $X_i \overset{\text{iid}}{\sim} \text{Beta}(\theta, 1)$, $\widehat{\theta} = -n/\sum_i \log X_i$ and $I_1(\theta) = 1/\theta^2$. Therefore

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} N(0, \theta^2).$$

Asymptotic Confidence Interval

Assuming the information $I_1(\theta)$ is continuous and positive,

$$\sqrt{nI_1(\widehat{\theta})}(\widehat{\theta} - \theta) \xrightarrow{d} N(0, 1).$$

An approximate $(1 - \alpha)100\%$ confidence interval is

$$\widehat{\theta} \pm \frac{z_{\alpha/2}}{\sqrt{nI_1(\widehat{\theta})}}.$$

Plug-in Fisher information estimates the unknown scale of the MLE. The interval's validity is asymptotic and depends on the regularity assumptions.

Example: Laplace MLE Confidence Interval

Example

Let $X_1, \ldots, X_n$ follow the Laplace location model $f(x \mid \theta) = \frac{1}{2}e^{-|x-\theta|}$. Construct a large-sample confidence interval from its MLE.

Solution. The MLE is a sample median $\widehat{\theta}_n$. For this model $I_1(\theta) = 1$, so

$$\sqrt{n}(\widehat{\theta}_n - \theta) \xrightarrow{d} N(0, 1).$$

An approximate $(1-\alpha)100\%$ confidence interval is

$$\widehat{\theta}_n \pm \frac{z_{\alpha/2}}{\sqrt{n}}.$$

Example

Let $X_i \overset{\text{iid}}{\sim} \text{Beta}(\theta, 1)$. Recall $\widehat{\theta} = -n/\sum_i \log X_i$ and $I_1(\theta) = 1/\theta^2$.

Solution.

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} N(0, \theta^2).$$

The plug-in information interval is

$$\widehat{\theta} \pm z_{\alpha/2} \frac{\widehat{\theta}}{\sqrt{n}}.$$

Corollary (Delta method for an MLE)

If g is differentiable at θ_0 with $g'(\theta_0) \neq 0$, the asymptotically normal MLE satisfies

$$\sqrt{n}\{g(\widehat{\theta}) - g(\theta_0)\} \xrightarrow{d} N\left(0, \frac{[g'(\theta_0)]^2}{I_1(\theta_0)}\right).$$

This is the delta method applied to the MLE. It complements the invariance property: invariance identifies the estimator, while the delta method gives its limiting distribution.

Sufficient Statistics

Data Reduction

Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$.

- Statistical inference starts with the full sample $X = (X_1, \ldots, X_n)$.
- We would like to summarize the observations without losing information about the parameter.
- A sufficient statistic makes that idea precise.

Definition (Sufficiency)

A statistic $T(X)$ is sufficient for θ if the conditional distribution of the full sample given $T = t$ does not depend on θ.

Example

Let $X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ and $T = \sum_i X_i \sim \text{Binomial}(n, \theta)$. Show that T is sufficient for θ.

Solution. For a binary sequence x containing exactly t successes,

$$P_\theta(X = x \mid T = t) = \frac{\theta^t(1-\theta)^{n-t}}{\binom{n}{t}\theta^t(1-\theta)^{n-t}} = \binom{n}{t}^{-1}.$$

The conditional distribution is independent of θ, so T is sufficient.

Theorem (Neyman–Fisher factorization)

For a dominated family with joint density or mass function $f(x \mid \theta)$, a statistic $T(X)$ is sufficient for θ if and only if

$$f(x \mid \theta) = k_1(T(x) \mid \theta) k_2(x),$$

where k_2 is independent of θ and both factors are nonnegative.

Proof. *Discrete case.* If T is sufficient, then, for $t = T(x)$,

$$P_\theta(X = x) = P_\theta(T = t) P(X = x \mid T = t),$$

which factorizes. Conversely, if $P_\theta(X = x) = k_1(t \mid \theta) k_2(x)$,

$$P_\theta(X = x \mid T = t) = \frac{k_2(x)}{\sum_{y:T(y)=t} k_2(y)}.$$

This conditional law is parameter-free. The same factorization result holds for dominated continuous models. □

Example

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Gamma}(2, \theta)$ using the scale parameter θ. Find a sufficient statistic for θ.

Solution.

$$f(x \mid \theta) = \theta^{-2n} \exp\left(-\frac{\sum_i x_i}{\theta}\right) \prod_{i=1}^{n} x_i.$$

The first factor depends on the data only through $T = \sum_i X_i$; the product $\prod_i x_i$ is parameter-free.

Example

Let X_i follow the shifted exponential density $f(x \mid \theta) = e^{-(x-\theta)}\mathbf{1}_{\{x>\theta\}}$. Find a sufficient statistic for θ.

Solution.

$$f(x \mid \theta) = e^{n\theta}\mathbf{1}_{\{\theta<x_{(1)}\}}\, e^{-\sum_i x_i}.$$

By factorization, $X_{(1)}$ is sufficient.

Example

Let $X_i \overset{\text{iid}}{\sim} N(\theta, \sigma_0^2)$ with σ_0^2 known. Show that $\bar{X}_n$ is sufficient for θ.

Solution. Since

$$\sum_i (x_i - \theta)^2 = \sum_i (x_i - \bar{x})^2 + n(\bar{x} - \theta)^2,$$

the joint density is a product of a factor depending on θ through $\bar{x}$ and a parameter-free factor.

Theorem (One-to-one transformation)

If T is sufficient for θ and h is one-to-one, then $h(T)$ is also sufficient for θ.

Theorem (MLE and sufficiency)

If an MLE exists uniquely, it can be expressed as a function of any sufficient statistic T.

The second statement follows from factorization: the parameter-free factor cannot affect the maximizer. It does not imply that every MLE is itself sufficient.

Theorem (Rao–Blackwell)

Let T be sufficient for θ, and let W be an unbiased estimator of $g(\theta)$ with finite variance. Then $\phi(T) = \mathrm{E}_\theta[W \mid T]$ is an estimator independent of the unknown θ, is unbiased for $g(\theta)$, and satisfies

$$\mathrm{Var}_\theta\{\phi(T)\} \leq \mathrm{Var}_\theta(W).$$

Proof. Sufficiency makes the conditional law used to define ϕ parameter-free. The tower property gives

$$\mathrm{E}_\theta[\phi(T)] = \mathrm{E}_\theta[\mathrm{E}_\theta[W \mid T]] = \mathrm{E}_\theta[W] = g(\theta).$$

The law of total variance yields

$$\mathrm{Var}_\theta(W) = \mathrm{Var}_\theta\{\phi(T)\} + \mathrm{E}_\theta[\mathrm{Var}_\theta(W \mid T)] \geq \mathrm{Var}_\theta\{\phi(T)\}.$$

□

Example

Let $X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$ and $T = \sum_i X_i$. Rao–Blackwellize the unbiased estimator $W = X_1$ by conditioning on T.

Solution.

$$\text{E}[W \mid T = t] = P(X_1 = 1 \mid T = t) = \frac{t}{n}.$$

Hence its Rao–Blackwellized form is

$$\text{E}[X_1 \mid T] = \frac{T}{n} = \bar{X}_n.$$

Definition (Minimal sufficient statistic)

A sufficient statistic T^* is minimal sufficient if, for any other sufficient statistic T, there is a function h with $T^* = h(T)$.

Theorem (Likelihood ratio criterion)

If the likelihood ratio $f(x \mid \theta)/f(y \mid \theta)$ is independent of θ exactly when $T(x) = T(y)$, then T is minimal sufficient, under the usual regularity of the criterion.

Theorem (Unique MLE criterion)

A unique MLE that is also sufficient is minimal sufficient.

Completeness and Uniqueness

Definition (Completeness)

A statistic T is complete for θ if, for every measurable g with an existing expectation,

$$\mathrm{E}_\theta[g(T)] = 0 \text{ for all } \theta \quad \Longrightarrow \quad P_\theta\{g(T) = 0\} = 1 \text{ for all } \theta.$$

Completeness guarantees uniqueness of an unbiased function of T, if such a function exists. It is a different requirement from sufficiency. Suppose $U_1(T)$ and $U_2(T)$ are both unbiased for the same $g(\theta)$. Then

$$\mathrm{E}_\theta[U_1(T) - U_2(T)] = 0 \qquad \text{for every } \theta.$$

Completeness forces

$$P_\theta\{U_1(T) = U_2(T)\} = 1 \qquad \text{for every } \theta.$$

Example

Let $T = \sum_i X_i \sim \text{Binomial}(n, \theta)$, $0 < \theta < 1$. Prove that T is complete.

Solution. If $\mathrm{E}_\theta[g(T)] = 0$ for all θ, then

$$0 = (1-\theta)^n \sum_{k=0}^{n} g(k) \binom{n}{k} \left(\frac{\theta}{1-\theta}\right)^k.$$

This polynomial is zero for every positive $\theta/(1-\theta)$, hence all its coefficients are zero: $g(k) = 0$ for $k = 0, \ldots, n$.

Example

Let $T = X_{(n)}$ from $X_i \overset{\text{iid}}{\sim} \text{Unif}(0, \theta)$, $\theta > 0$. Prove that T is complete.

Solution. Its density is $f_T(t) = nt^{n-1}/\theta^n$ for $0 < t < \theta$. If $\text{E}_\theta[g(T)] = 0$ for every $\theta > 0$, then

$$\int_0^\theta g(t)t^{n-1}\,dt = 0 \qquad \text{for all } \theta > 0.$$

Differentiation gives $g(\theta)\theta^{n-1} = 0$ almost everywhere.

Definition (Complete sufficient statistic)

A statistic that is both complete and sufficient for θ is called a complete sufficient statistic (CSS).

Theorem

A one-to-one transformation of a CSS is again a CSS.

Theorem

A complete sufficient statistic is also minimal sufficient (up to almost-sure equivalence).

Definition (Regular exponential class)

A family of distributions belongs to the regular exponential class if its pmf/pdf can be expressed as

$$f(x \mid \theta) = \exp\{p(\theta)T(x) + H(x) + q(\theta)\}, \qquad x \in S,$$

where $\theta \in \Omega = (\gamma, \delta)$ with $\gamma < \delta$, and the following conditions hold:

(i) The support S does not depend on θ.

(ii) $p(\theta)$ is a nonconstant continuous function of θ.

(iii) If X is continuous, $T'(x) \neq 0$ and $H(x)$ is continuous on S. If X is discrete, $T(x)$ is a nonconstant function on S.

Example

Let $X \sim N(0, \theta)$ with $\theta > 0$. Express its density in exponential-class form.

Solution.

$$f(x \mid \theta) = \exp\left\{-\frac{x^2}{2\theta} - \frac{1}{2}\log(2\pi\theta)\right\}.$$

The exponential-class components can be taken as

$$p(\theta) = -\frac{1}{2\theta}, \quad T(x) = x^2, \quad H(x) = 0, \quad q(\theta) = -\tfrac{1}{2}\log(2\pi\theta).$$

Example

Let $X \sim \text{Binomial}(m, \theta)$, $0 < \theta < 1$. Express its pmf in exponential-class form.

Solution.

$$P_\theta(X = x) = \exp\left\{x \log \frac{\theta}{1-\theta} + m \log(1-\theta) + \log \binom{m}{x}\right\}.$$

Thus $T(x) = x$ and $p(\theta) = \log\{\theta/(1-\theta)\}$.

Complete Sufficiency in the Exponential Class

For an i.i.d. sample from the regular exponential class,

$$f(x \mid \theta) = \exp\left\{ p(\theta) \sum_{i=1}^{n} T(x_i) + \sum_{i=1}^{n} H(x_i) + nq(\theta) \right\}.$$

By factorization, $\sum_i T(X_i)$ is sufficient. For a full regular one-parameter exponential family with an open natural parameter space, it is also complete.

This is a practical way to obtain a CSS without proving completeness by a polynomial or integral argument in every example.

Definition (MVUE)

An unbiased estimator T^* of θ is an MVUE if for every other unbiased estimator T,

$$\mathrm{Var}_\theta(T^*) \leq \mathrm{Var}_\theta(T) \qquad \text{for all } \theta \in \Theta.$$

Theorem (Lehmann–Scheffé)

Let T be complete and sufficient for θ. If $\phi(T)$ has finite variance and is unbiased for $g(\theta)$, then $\phi(T)$ is the unique minimum-variance unbiased estimator of $g(\theta)$, up to almost-sure equality.

Proof. For any unbiased estimator U of $g(\theta)$, Rao–Blackwell gives

$$\mathrm{Var}_\theta\{\mathrm{E}[U \mid T]\} \le \mathrm{Var}_\theta(U).$$

Both $\mathrm{E}[U \mid T]$ and $\phi(T)$ are unbiased functions of T. By completeness they coincide almost surely. Therefore

$$\mathrm{Var}_\theta\{\phi(T)\} \le \mathrm{Var}_\theta(U) \qquad \text{for every } \theta.$$

Completeness also gives uniqueness among unbiased functions of T, hence uniqueness of the MVUE. □

How to Find the MVUE

Three useful approaches are:

1. Show an unbiased estimator attains the CRLB.
2. Start with an unbiased estimator U and compute $\mathrm{E}[U \mid T]$ for a CSS T.
3. Find an unbiased function $\phi(T)$ of a CSS directly.

Example

Let $X_i \overset{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ with $n \geq 2$ and $T = \sum_i X_i$ a CSS. Find MVUEs of θ and θ^2.

Solution. Since $\bar{X}_n = T/n$ is unbiased for θ, it is the MVUE of θ. For θ^2, start with $U = X_1 X_2$, which is unbiased. Then

$$\text{E}[X_1 X_2 \mid T = t] = \frac{t(t-1)}{n(n-1)}.$$

Hence $T(T-1)/[n(n-1)]$ is the MVUE of θ^2 by Lehmann–Scheffé.

Example

Let $X_i \overset{\text{iid}}{\sim} \text{Unif}(0, \theta)$, $\theta > 0$. The sample maximum $T = X_{(n)}$ is complete and sufficient. Find the MVUE of θ.

Solution. Its expectation is

$$\text{E}_\theta[T] = \frac{n}{n+1}\theta.$$

Thus the unbiased function

$$\widehat{\theta}_{\text{MVUE}} = \frac{n+1}{n} X_{(n)}$$

is the unique MVUE of θ.

Example

Let $X_i \overset{\text{iid}}{\sim} N(\theta, \sigma_0^2)$ with known σ_0^2. The sample mean $\bar{X}_n$ is a CSS for θ. Find MVUEs of θ and θ^2.

Solution. Since $\mathrm{E}_\theta[\bar{X}_n] = \theta$, the MVUE of θ is $\bar{X}_n$. Also,

$$\mathrm{E}_\theta[\bar{X}_n^2] = \theta^2 + \frac{\sigma_0^2}{n}.$$

Thus $\bar{X}_n^2 - \sigma_0^2/n$ is the MVUE of θ^2.

Appendix: Proofs

Proof of the Score Identities

Proof. Since the support does not depend on θ and differentiation under the integral is permitted,

$$\int f(x \mid \theta)\, dx = 1, \qquad \int \frac{d}{d\theta} f(x \mid \theta)\, dx = 0.$$

At points where $f(x \mid \theta) > 0$,

$$\frac{d}{d\theta} \log f(x \mid \theta) = \frac{\frac{d}{d\theta} f(x \mid \theta)}{f(x \mid \theta)}.$$

Therefore,

$$\mathrm{E}_\theta[\frac{d}{d\theta} \log f(X \mid \theta)] = \int \frac{\frac{d}{d\theta} f(x \mid \theta)}{f(x \mid \theta)} f(x \mid \theta)\, dx = 0.$$

The same calculation applies to probability mass functions, with sums replacing integrals.

Proof (continued). Differentiating the identity for the log-density once more,

$$\frac{d^2}{d\theta^2}\log f(x \mid \theta) = \frac{\frac{d^2}{d\theta^2} f(x \mid \theta)}{f(x \mid \theta)} - \left(\frac{\frac{d}{d\theta} f(x \mid \theta)}{f(x \mid \theta)}\right)^2.$$

Multiplying by $f(x \mid \theta)$, integrating and using the second-order interchange condition give

$$\mathrm{E}_\theta[\frac{d^2}{d\theta^2}\log f(X \mid \theta)] = \frac{d^2}{d\theta^2}\int f(x \mid \theta)\, dx - \mathrm{E}_\theta[(\frac{d}{d\theta}\log f(X \mid \theta))^2].$$

Since the integral equals 1 for all θ, its second derivative is 0. Consequently,

$$\mathrm{E}_\theta[(\frac{d}{d\theta}\log f(X \mid \theta))^2] = -\mathrm{E}_\theta[\frac{d^2}{d\theta^2}\log f(X \mid \theta)].$$

Proof. Under the theorem's assumptions, the sample log-likelihood is a sum of individual log-densities, and hence

$$\ell_n'(\theta) = \sum_{i=1}^{n} \frac{d}{d\theta} \log f(X_i \mid \theta).$$

By the first score identity, each summand has mean zero. Thus

$$\mathrm{E}_\theta[\ell_n'(\theta)] = \sum_{i=1}^{n} \mathrm{E}_\theta[\frac{d}{d\theta} \log f(X_i \mid \theta)] = 0.$$

Because the observations are independent, the individual score terms are independent. Their variances therefore add:

$$\mathrm{Var}_\theta(\ell_n'(\theta)) = \sum_{i=1}^{n} \mathrm{Var}_\theta(\frac{d}{d\theta} \log f(X_i \mid \theta)) = nI_1(\theta).$$

Proof (continued). Since $\mathrm{E}_\theta[\ell_n'(\theta)] = 0$,

$$\mathrm{Cov}_\theta(Y, \ell_n'(\theta)) = \mathrm{E}_\theta[Y\ell_n'(\theta)].$$

For the joint density of an i.i.d. sample, the product rule yields

$$\frac{d}{d\theta}\prod_{i=1}^{n} f(x_i \mid \theta) = \left(\prod_{i=1}^{n} f(x_i \mid \theta)\right)\sum_{i=1}^{n}\frac{d}{d\theta}\log f(x_i \mid \theta).$$

Consequently, the covariance may be written as

$$\mathrm{Cov}_\theta(Y, \ell_n'(\theta)) = \int Y(x_1, \ldots, x_n)\frac{d}{d\theta}\left\{\prod_{i=1}^{n} f(x_i \mid \theta)\right\} dx_1 \cdots dx_n.$$

Proof (continued). The regularity assumption permits differentiation under the integral for the integrand weighted by Y. Thus

$$\mathrm{Cov}_\theta(Y, \ell_n'(\theta)) = \frac{d}{d\theta} \int Y(x_1, \ldots, x_n) \prod_{i=1}^{n} f(x_i \mid \theta)\, dx_1 \cdots dx_n.$$

The integral on the right is $\mathrm{E}_\theta[Y] = k(\theta)$. We conclude that

$$\mathrm{Cov}_\theta(Y, \ell_n'(\theta)) = \frac{d}{d\theta} \mathrm{E}_\theta[Y] = k'(\theta).$$

One sufficient condition for the interchange is an integrable bound, valid throughout a neighborhood of θ, on the absolute value of the differentiated joint density multiplied by $|Y|$. In a discrete model, use sums instead of integrals.

Proof (continued). Cauchy–Schwarz for the centered random variables $Y - \mathrm{E}_\theta[Y]$ and $\ell_n'(\theta)$ gives

$$\{\mathrm{Cov}_\theta(Y, \ell_n'(\theta))\}^2 \leq \mathrm{Var}_\theta(Y)\,\mathrm{Var}_\theta(\ell_n'(\theta)).$$

Substituting the identities established above,

$$\{k'(\theta)\}^2 \leq \mathrm{Var}_\theta(Y) n I_1(\theta).$$

Since $0 < I_1(\theta) < \infty$, division gives

$$\mathrm{Var}_\theta(Y) \geq \frac{\{k'(\theta)\}^2}{n I_1(\theta)}.$$

Equality holds precisely when the centered estimator is a constant multiple of the score, namely

$$Y - \mathrm{E}_\theta[Y] = \frac{k'(\theta)}{n I_1(\theta)} \ell_n'(\theta) \quad P_\theta\text{-a.s.}$$

Score and Information Limits Used in the Proof

Under the score identities and finite positive Fisher information,

$$\mathrm{E}_{\theta_0}[\frac{d}{d\theta}\log f(X \mid \theta_0)] = 0, \qquad \mathrm{Var}_{\theta_0}(\frac{d}{d\theta}\log f(X \mid \theta_0)) = I_1(\theta_0).$$

By the central limit theorem,

$$\frac{\ell_n'(\theta_0)}{\sqrt{n}} \xrightarrow{d} N(0, I_1(\theta_0)).$$

The weak law of large numbers and suitable smoothness give

$$-\frac{\ell_n''(\widetilde{\theta}_n)}{n} \xrightarrow{P} I_1(\theta_0).$$

These two limits, combined with the Taylor expansion of the likelihood equation and Slutsky's theorem, imply asymptotic normality of the MLE.

Proof of the Asymptotic Normality of the MLE

Proof. Under the regularity conditions of the stated theorem, Taylor's theorem applied to the score at the consistent interior MLE gives

$$0 = \ell_n'(\widehat{\theta}_n) = \ell_n'(\theta_0) + (\widehat{\theta}_n - \theta_0)\ell_n''(\widetilde{\theta}_n),$$

where $\widetilde{\theta}_n$ lies between $\widehat{\theta}_n$ and θ_0. Consequently,

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = \frac{n^{-1/2}\ell_n'(\theta_0)}{-n^{-1}\ell_n''(\widetilde{\theta}_n)}.$$

The numerator obeys the central limit theorem,

$$\frac{\ell_n'(\theta_0)}{\sqrt{n}} \xrightarrow{d} N(0, I_1(\theta_0)).$$

Proof (continued). By consistency, $\widetilde{\theta}_n \xrightarrow{P} \theta_0$. The regularity and integrability assumptions, together with the WLLN, yield

$$-\frac{\ell_n''(\widetilde{\theta}_n)}{n} \xrightarrow{P} I_1(\theta_0) > 0.$$

The denominator is therefore nonzero with probability tending to one. Applying Slutsky's theorem to the ratio,

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \xrightarrow{d} N\left(0, \frac{1}{I_1(\theta_0)}\right).$$

Thus the asserted limiting distribution follows. □

Proof of the One-Step Estimation Theorem

Proof. Because the initial estimator has error of order $n^{-1/2}$, Taylor expansion of the score around θ gives

$$\ell_n'(\widehat{\theta}_n^{(0)}) = \ell_n'(\theta) + \ell_n''(\theta)(\widehat{\theta}_n^{(0)} - \theta) + o_P(\sqrt{n}).$$

Under the stated regularity conditions, the normalized second derivatives obey

$$-\frac{\ell_n''(\theta)}{n} \xrightarrow{P} I_1(\theta), \qquad -\frac{\ell_n''(\widehat{\theta}_n^{(0)})}{n} \xrightarrow{P} I_1(\theta).$$

Substitute the expansion into the one-step update. To first order, the correction cancels the initial estimation error, leaving

$$\sqrt{n}(\widehat{\theta}_n^{(1)} - \theta) = \frac{\ell_n'(\theta)/\sqrt{n}}{I_1(\theta)} + o_P(1).$$

Proof (continued). By the score identities,

$$\mathrm{E}_\theta[\ell_n'(\theta)] = 0, \qquad \mathrm{Var}_\theta(\ell_n'(\theta)) = nI_1(\theta).$$

Therefore the central limit theorem gives

$$\frac{\ell_n'(\theta)}{\sqrt{n}} \xrightarrow{d} N(0, I_1(\theta)).$$

Combining this with the preceding expansion and applying Slutsky's theorem,

$$\sqrt{n}(\widehat{\theta}_n^{(1)} - \theta) \xrightarrow{d} N\left(0, \frac{1}{I_1(\theta)}\right).$$

Thus one Newton correction has the same first-order limiting distribution as the regular MLE. □

References

References

- Lee, K. (2024). *Introduction to Statistical Inference [Lecture notes]*. Statistics. Sungkyunkwan University.
- 김우철, *수리통계학* (개정판), 민음사, 제6장 “추정”.

Thank you!